

REPORT OF USED CARS PRICE ANALYSIS

1. Declaration

I hereby declare that this project titled “Used Cars Price Analysis – Multi-City Market Insights” is an original work carried out by me and has not been submitted previously for any academic purpose.

2. Abstract

This project focuses on analyzing used car listings collected from multiple Indian cities to understand pricing trends, brand performance, and market segmentation.

The data was collected through web scraping from the CarDekho website using Python and Selenium. After collecting the data city-wise, it was cleaned, consolidated, and transformed using Power BI.

Exploratory Data Analysis (EDA) was conducted to understand data distribution and pricing patterns. Finally, an interactive three-page dashboard was created to provide meaningful insights into the used car market.

The analysis highlights how brand, fuel type, mileage, and city influence vehicle pricing.

3. Introduction

The used car market in India has grown significantly due to increasing affordability and digital automobile platforms. Consumers rely heavily on online platforms to compare pricing and vehicle details.

However, pricing variations across brands, fuel types, transmission modes, and cities are not always easy to interpret.

This project aims to analyze used car data across multiple cities to identify key market trends and pricing behavior using data analytics tools.

4. Problem Statement

The used car market involves multiple influencing factors including brand reputation, manufacturing year, mileage, fuel type, and geographic location.

The objective of this project is to:

- Analyze price variation across brands
 - Identify dominant cities in used car listings
 - Examine the impact of fuel type and transmission on pricing
 - Study the relationship between kilometers driven and resale value
 - Build interactive dashboards for multi-city comparison
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5. Project Objectives

- Collect used car data through web scraping
 - Consolidate city-wise datasets
 - Clean and standardize raw data
 - Perform exploratory data analysis
 - Design interactive Power BI dashboards
 - Extract actionable market insights
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6. Methodology

The project followed a structured workflow:

1. Data Collection (Web Scraping)
2. Data Consolidation (Appending City Files)
3. Data Cleaning and Transformation
4. Exploratory Data Analysis
5. Dashboard Design and Visualization
6. Insight Extraction and Reporting

This step-by-step approach ensured accurate and meaningful analysis.

7. Data Collection

Data was collected from the CarDekho website using Python, Selenium, and BeautifulSoup.

Process:

- Automated browser navigation
- Handled pagination
- Extracted relevant fields
- Stored data city-wise

Fields Collected:

- Car Brand
- Car Name
- Year

- Kilometers Driven
- Fuel Type
- Transmission
- Price
- City

Data was collected for 13 major Indian cities and stored in CSV format.

8. Data Cleaning and Preparation

8.1 Data Consolidation

All city-wise CSV files were appended into a single consolidated dataset to enable multi-city analysis.

8.2 Removal of Missing and Duplicate Records

Blank and duplicate records were removed to ensure accurate counts.

8.3 Price Standardization

Price values were cleaned and converted into numeric INR format for accurate calculations.

8.4 Text Standardization

Fuel type, transmission, city, and brand names were standardized to maintain consistency.

8.5 Derived Columns Created

- Price Range
- KM Range
- Car Age (if applicable)

8.6 Data Type Validation

All columns were converted to appropriate numeric or text data types.

9. Exploratory Data Analysis (EDA)

EDA was conducted to understand patterns and distribution before dashboard development.

Key Observations:

- Most cars fall within ₹2–10 Lakhs price range
- Petrol vehicles dominate listings
- Manual transmission vehicles are more common

- Lower mileage vehicles have higher resale value
- Recent manufacturing years show higher listing volume

EDA guided the structure of the dashboard.

10. Dashboard Design and Implementation

A three-page interactive Power BI dashboard was developed.

Page 1: Market Overview

Purpose: Provide high-level summary.

Includes:

- Total Cars Listed
 - Average Car Price
 - Cities Covered
 - Most Listed Brand
 - Cars by City
 - Price Range Distribution
 - Top Brands by Listings
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Page 2: Price & Brand Analysis

Purpose: Analyze pricing behavior.

Includes:

- Average Price by Brand
 - Average Price by Fuel Type
 - Average Price by KM Range
 - Price Trend by Year
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Page 3: Best Value Analysis

Purpose: Understand market segmentation.

Includes:

- Top brands in 5-10 lakhs segment
- Average km_driven by price_range

- Top 10 best value cars

Slicers were added for interactive filtering.

11. Key Insights

- Maruti is the most frequently listed brand
 - Majority of cars fall in the mid-range price category
 - Petrol and manual transmission dominate the market
 - Mileage significantly impacts resale price
 - Luxury brands have higher average pricing
 - City-wise variations exist in listing volume
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12. Challenges Faced

- Handling dynamic web elements during scraping
 - Managing pagination across multiple pages
 - Converting textual price formats into numeric values
 - Standardizing inconsistent text values
 - Managing large multi-city dataset
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13. Conclusion

This project successfully demonstrated an end-to-end data analytics workflow, from web scraping to dashboard visualization.

The analysis provided valuable insights into pricing trends, brand performance, and market segmentation in the Indian used car market.

The interactive dashboard enables dynamic comparison across cities and vehicle attributes, making it a practical analytical tool.

14. Future Scope

- Include additional cities for broader analysis
- Perform predictive price modeling using machine learning
- Integrate EMI and financing analysis
- Compare multiple automobile platforms